

Generalized Quantum Signal Processing for Hamiltonian Simulation

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Abstract — This proposal aims for the implementation of the complex time evolution of a given Hamiltonian $\hat{H}$ using the recent technique of Generalized Quantum Signal Processing (GQSP). In this work, we will present an introduction to the concept of QSP protocol using rotation around R_z and R_x gate, and how this can be generalized for any gate over $SU(2)$. We will study how the accuracy of the simulation behaves in relation with the degree of the polynomial. If the time is permitted, we will explore how different polynomial approximation can be achieved in relation with the system to simulate.

1. INTRODUCTION

Hamiltonian simulation using real quantum computers is one of the ideas envisioned by Feynman back in 1981 [1]. With the rapid development of modern quantum hardware—such as superconducting circuits [2], neutral-atoms [3], photonic platforms [4], and trapped ions [5], among others—there is a growing need for algorithms capable of performing accurate and efficient quantum time evolution for applications in chemistry, materials science, and many-body physics. Classical approximations for quantum dynamics scale poorly with system size, making quantum algorithms an appealing route for studying molecular reactions. However, the usefulness of quantum devices depends on the availability of simulation techniques that combine low gate complexity with controllable approximation error.

In this context, Quantum Signal Processing (QSP) [6] and its generalization have emerged as promising frameworks for implementing efficient polynomial transformations of unitary operators. Nonetheless, standard QSP imposes structural constraints—particularly parity restrictions that limit its applicability for simulating general Hamiltonian evolutions. Generalized Quantum Signal Processing (GQSP) [7] removes these constraints by incorporating general $SU(2)$ rotations, enabling greater expressivity and more flexible polynomial constructions.

In this paper, we investigate the use of GQSP for Hamiltonian simulation in the context of chemical reaction dynamics. We validate the method numerically, and apply it to the three-hydrogen system H_3 , analyzing how the accuracy of the GQSP approximation depends on the polynomial degree and the underlying geometry of the molecular configuration.

2. EXISTING LITERATURE REVIEW

Early quantum simulation proposals focused on digital approximations of quantum time evolution. Among the earliest are the Trotter-Suzuki product formulas [8], which decompose the unitary evolution operator e^{-iHt} into products of exponentials of individual Hamiltonian terms. These formulas provide approximations whose accuracy improves with higher-order decompositions. However, their gate complexity grows as $O(t^2/\epsilon)$, making them costly for long evolution times or Hamiltonians composed of many non-commuting components. Beyond product formulas, block-encoding techniques led to alternatives such as the Linear Combination of Unitaries (LCU) method [9], which approximates the time-evolution operator using truncated Taylor expansions embedded into controlled unitaries. Although LCU achieves improved asymptotic behavior, it requires amplitude amplification, ancilla qubits, and controlled block-encodings, contributing to significant overhead.

A major conceptual advance occurred with the development

of Quantum Signal Processing (QSP) [6] $P(x)$ using alternating applications of fixed signal operator and adjustable single-qubit rotations. Its generalization, Quantum Singular Value Transformation (QSVT) [6], extends this idea to arbitrary block-encoded operators, providing a unified framework for Hamiltonian simulation, linear algebra algorithms, and amplitude amplification. QSP and QSVT achieve near-optimal complexity, scaling polylogarithmically in $1/\epsilon$, which represents a substantial improvement over Trotter-based approaches.

Despite their efficiency, standard QSP imposes structural parity constraints on the implementable polynomials $P(x)$ and $Q(x)$, limiting the set of functions such as e^{-iHt} that lack definite parity.

Generalized Quantum Signal Processing (GQSP) [7] overcomes these limitations by allowing arbitrary $SU(2)$ rotations between signal-processing steps, removing parity restrictions and broadening the class of achievable polynomial transformations. Extensions of QSP and GQSP have also been developed for continuous-variables systems [10] and hybrid quantum architectures [11], positioning GQSP as a promising approach for molecular simulation and chemical reaction dynamics.

3. PROBLEM FORMULATION

Hydrogen chains and small hydrogen clusters are used as toy model for benchmarking quantum simulation techniques due to their simplicity and the rich physical behavior they exhibit. In this work, we focus on the H_3 system, which provides a minimal setting to study reaction-like dynamics, population transfer, and the performance of Hamiltonian simulation methods. We consider a geometry where the two outer hydrogen atoms are fixed at the equilibrium bond distance of H_2 , while the central hydrogen atom is displaced along the molecular axis. This one-dimensional motion generates a characteristic double-well potential: two symmetric minima corresponding to the dissociation limit $H_2 + H$ and a central high-energy region dominated by Coulomb repulsion. These features make the system suitable for testing the accuracy of polynomial-approximation-based simulation methods.

Along this reaction coordinate, we select two representative configurations:

- **Equilibrium point ($H_2 + H$):** the central atom lies near one of the low-energy minima of the potential.
- **Non-equilibrium point ($H + H + H$):** the central atom is positioned near the high-energy repulsive region.

To prove the system’s dynamics, we study the time evolution of an initial electronic configuration $|\psi_0\rangle = |010110\rangle$ and non-

itor population transfer on the fifth site through the occupation operator $\hat{n}_5 = \sigma_- \sigma_+$.

This observable allows us to track how electronic population redistributes during the evolution across the two selected geometries. Our computational task is to construct, using QSP a polynomial approximation to the time-evolution operator $e^{-i\hat{H}t}$ corresponding to each configuration. We then analyze how the quality of the QSP simulation depends on the polynomial order and evaluate its accuracy across both geometries.

Figure 1 illustrates the potential energy surface of the H_3 system as a function of the displacement of the central hydrogen atom. The left panel shows the full-three dimensional landscape, highlighting the two symmetric minima associated with the dissociation channels $H_2 + H$. The right panel provides a top view of the surface together with the reaction coordinate (red line) along which the QSP simulations are performed. The blue and green markers indicate the non-equilibrium and equilibrium configurations studied in this work.

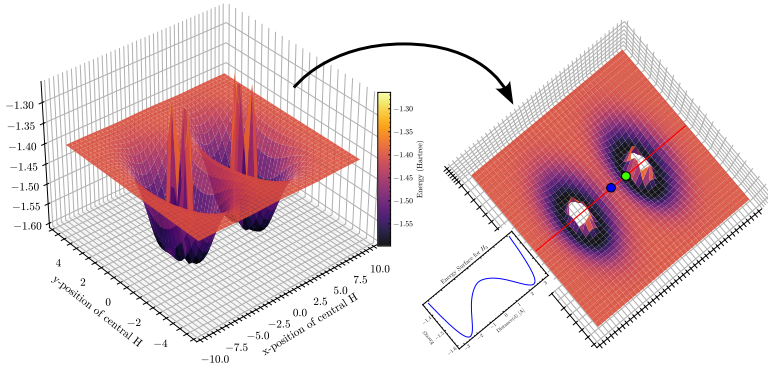


Figure 1: Potential energy landscape of the H_3 system. The left panel shows the three-dimensional surface, while the right panel provides a top view including the reaction coordinate and the two selected geometries (blue: non-equilibrium, green: equilibrium).

4. METHODOLOGY

4.1 Quantum Signal Processing

Early experiments in NMR shows the importance of the BB1 protocol for filtering. The same idea is applied to quantum circuits for generating a sequence of non-commuting gates to approximate a required unitary operation. QSP construct a polynomial transformation of the signal operator

$$W(x) = \begin{pmatrix} x & i\sqrt{1-x^2} \\ -i\sqrt{1-x^2} & x \end{pmatrix} \quad (1)$$

together with single-qubit R_z rotations,

$$R_z(\theta) = \begin{pmatrix} \exp(-i\theta) & 0 \\ 0 & \exp(i\theta) \end{pmatrix} \quad (2)$$

The signal operation can be implemented through a controlled X rotation with angle $\phi = -2 \arccos(x)$. By alternating $W(x)$ with the sequence of phases $\theta \in (\theta_0, \dots, \theta_{d-1})$, QSP realizes a unitary of the form

$$U_\theta = R_z(\theta_0) \prod_{i=1}^{d-1} W(x) R_z(\theta_i) \\ = \begin{pmatrix} P(x) & iQ(x)\sqrt{1-x^2} \\ iQ(x)\sqrt{1-x^2} & P(x) \end{pmatrix}. \quad (3)$$

This implementation must satisfy three feasibility constraints:

1. The degree of the polynomial is $P(x) \leq d - 1$.
2. The function has parity $d \bmod 2$.
3. $|P(x)|^2 + (1 - x^2)|Q(x)|^2 = 1$.

A crucial component of QSP is the synthesis of the phases $\{\theta_k\}$ required to implement a desired polynomial. Several efficient algorithms exist for angle synthesis; however, the parity constraint constitutes a significant limitation for Hamiltonian simulation. For example, to simulate $\exp(-i\hat{H}t)$, where $\hat{H}$ is my Hamiltonian of interest. As the exponential function does not have a defined parity we need to decompose the function into two different parity functions using the Euler's identity.

$$\exp(i\hat{H}t) = \cos(\hat{H}t) + i \sin(\hat{H}t) \quad (4)$$

Thus, standard QSP requires separate implementations of $\cos(\cdot)$ and $\sin(\cdot)$, followed by a Linear Combination of Unitaries (LCU) procedure[9] to recombine the two parts-doubling the circuit depth and complicating control logic.

4.2 Generalized Quantum Signal Processing

To overcome the parity constraints of standard QSP, Generalized QSP extends the allowed interleaving gates from $\{R_x, R_z\}$ to the full space of $SU(2)$ rotations. A general single-qubit $SU(2)$ gate may be written as

$$U(\lambda, \theta, \phi) = \begin{pmatrix} e^{i(\lambda+\phi)} \cos \theta & e^{i\phi} \sin \theta \\ e^{i\lambda} \sin \theta & -\cos \theta \end{pmatrix}. \quad (5)$$

This expression allows us to bypass the parity constraint from the original QSP. Fig. 2-b shows the circuit diagram of the generalized QSP.

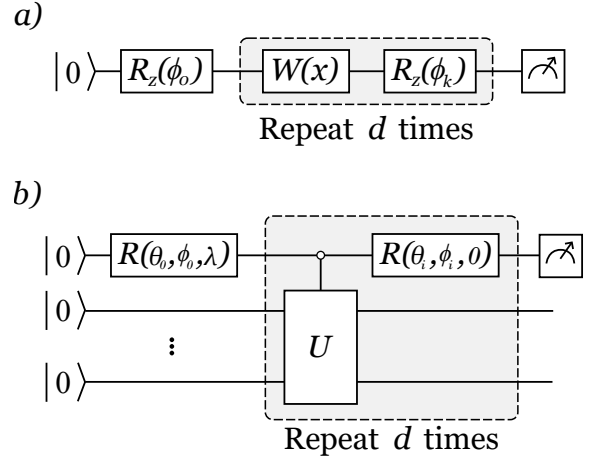


Figure 2: a) Quantum Signal Processing circuit representation. b) Generalized Quantum Signal Processing circuit representation.

4.2.1 Classical Brute Force to Find Angles

To simulate a generalized QSP block, it is important to find the angles λ, θ and ϕ . The number of angles to find depends on the degree d of the polynomial to be implemented. The classical method is two folds- 1. Compute the target matrix T . 2. Define the loss function $\mathcal{L}$ for the optimizer. The target matrix is found using the Hamiltonian equation $e^{-i\hat{H}t}$ where H is the required unitary matrix. $e^{-i\hat{H}t}$ is expanded using Taylor's expansion with

power d . GQSP matrix is defined by matrix multiplication of Fig 1.b. Loss function is defined as

$$\mathcal{L} = \| \|U_{00}\| - \|T\| \| \quad (6)$$

This loss function is passed to the COBYLA optimizer to find the angles. We plotted the time taken to get the angles for degree $d = 20$, with size of matrix n from 2×2 to 32×32 and time in the Hamiltonian simulation $t \in [0, 20]$ with the error on loss function to 1%.

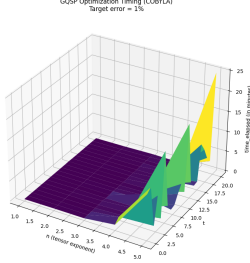


Figure 3: Classical Brute Force simulation time

The time complexity for simulation is $\mathcal{O}(n^3 d^2)$. The computations got expensive with every increase of n and d making it almost impossible to classically find the angles and using them for GQSP.

4.3 Block-encoding for Hamiltonian simulation

While very practical, we need to consider that most cases the system of interest can not be implemented as a unitary gate. But rather we need to block-encode the given unitary. This transformation change the polynomial transformation that we need to implement in the GQSP framework. In order to illustrate this, consider the following block-encode of H .

$$U_H = \begin{pmatrix} \frac{H}{i\sqrt{I - HH^\dagger}} & i\sqrt{I - HH^\dagger} \\ & H^\dagger \end{pmatrix} \quad (7)$$

If we work in the eigen-basis of H , and recalculate the eigenvalues of U_H the new eigenvalues are $0.5 * (\lambda - \lambda^{-1})$. We need to implement the following polynomial transformation for a block-encoding as in matrix 7.

$$\tilde{U} = \exp(-it(H + H^{-1})/2) \quad (8)$$

4.4 Polynomial construction and Q -complement

Given a target polynomial $P(x)$ we compute the complementary polynomial $Q(x)$ required to satisfy the unitary condition using Fejér-Riesz decomposition. Our numerical implementation constructs the loss function via FFT-based convolution and optimizes coefficients using L-BFGS until the constraint $|P|^2 + |Q|^2 = 1$ is satisfied within numerical tolerance.

4.5 Angle extraction via inverse recurrence

The GQSP angles $(\lambda, \theta_k, \phi_k)$ are obtained by running the generalized QSP recurrence relation in reverse:

$$(P_k, Q_k) = A^\dagger R_k^\dagger (P_{k+1}, Q_{k+1}) \quad (9)$$

and extracting

$$\theta_k = \frac{1}{2} \arccos(\cdot), \quad \phi_k = \arg(\cdot). \quad (10)$$

This procedure is repeated until reaching grade zero, yielding the full GQSP sequence.

5. RESULTS

5.1 Validation of the GQSP Implementation (polynomial tests)

To verify the correctness and numerical behavior of the GQSP implementation, we tested three different polynomial pairs (P, Q) and compared the resulting GQSP block $F_{\text{GQSP}} = \langle 0|U_{\text{GQSP}}|0\rangle$ against the target function $F(x)$.

Case	$P(x)$	Q obtained	$(P ^2 + Q ^2)$	$ F(x) - F_{\text{GQSP}} $
1	$\frac{1}{4}(x^2 + 3x)$	$\frac{\sqrt{3}}{4}(-x^2 + x)$	1.00/1.00	2.78×10^{-16}
2	$\frac{1}{4}(x^2 + 3x)$	FFT + Fejér-Riesz	0.999/1.00	2.25×10^{-16}
3	Taylor e^{-6ix} (L=20)	FFT + Fejér-Riesz	0.99/1.00	4.55×10^{-6}

Table 1: Comparison between the target operator $F(x)$ and the implemented block F_{GQSP} for three different polynomial constructions.

5.2 Time Evolution Under GQSP

Figure 4 shows the simulated dynamics of the occupation number on the fifth qubit, for polynomial orders 30, 60, and 90. The initial state is $|\psi_0\rangle = |010110\rangle$ and the observable $\hat{n}_5 = \sigma_- \sigma_+$ tracks population transfer across the two configurations. In the **equilibrium configuration** (Fig. 4a), all orders of GQSP qualitatively reproduce the oscillatory behavior of the exact evolution. Increasing the polynomial degree improves the quantitative match, with order 90 nearly overlapping the exact curve across the full simulation window. In the **non-equilibrium configuration** (Fig. 4b), the dynamics exhibit stronger variation due to the deeper local potential. Lower-degree approximation deviate more noticeably from the exact trajectory, whereas higher-degree GQSP (order 90) captures both the amplitude and phase of the improved accuracy. These results demonstrate that GQSP provides a controllable and accurate approximation of the molecular time evolution, with performance improving systematically as the polynomial degree increases.

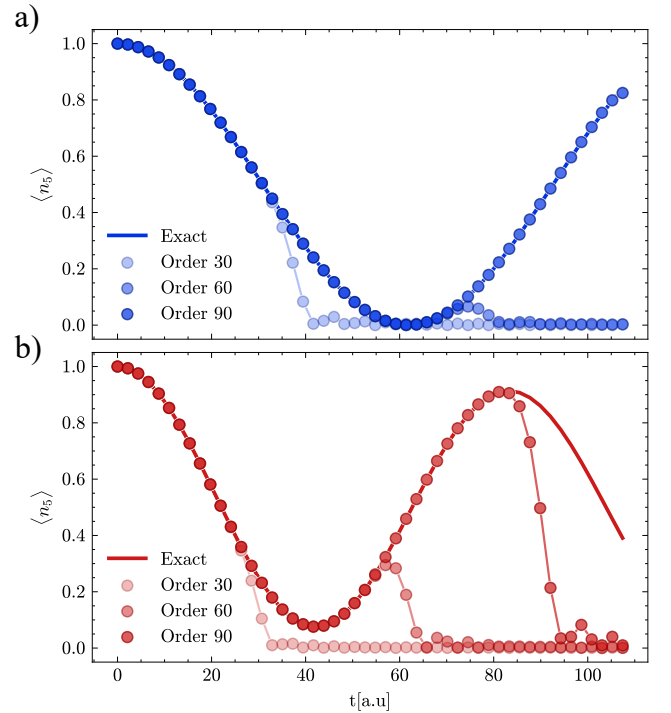


Figure 4: Generalized Quantum Signal processing result for chemical reactions of H_3 . a) Equilibrium energy of interaction $H + H_2$ under evolution. b) Non-equilibrium point of interaction $H + H + H$ under evolution.

6. DISCUSSION

Our results demonstrate that GQSP provides a flexible and accurate framework for Hamiltonian simulation beyond the constraint of standard QSP. Its ability to implement general polynomial transformations avoids the parity limitations that complicate simulating functions like e^{-iHt} . For the H_3 reaction model, we find that GQSP reproduces both the qualitative and quantitative features of the exact dynamics, and the accuracy improves systematically with the polynomial degree.

In the equilibrium geometry, even polynomial orders (e.g., 60) capture the oscillatory behavior with high fidelity, where the non-equilibrium configuration requires higher-degree polynomials due to sharper features of crucial potential energy surface. This highlights an important characteristic of polynomial-based Hamiltonian simulation: system with stronger local curvature or stronger non-commuting terms demand higher approximation order. Extending this framework to larger molecules and multidimensional reaction coordinates represents an important next step toward assessing the scalability of GQSP for realistic quantum chemistry applications.

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